

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 October 2022 Worked Solutions

October 2022 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P2

PAST PAPER

WMA12/01 October 2022

October 2022 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Proof

1. Given that a , b and c are integers greater than 0 such that

- $c = b + 2$
- $a + b + c = 10$

Prove, by exhaustion, that the product of a , b and c is always even.

You may use the table below to illustrate your answer.

(3)

You may not need to use all rows of this table.

a	b	c
	1	
	2	

Worked Solution - Question 1

Topic group

1. Use the two conditions

Since $c = b + 2$ and $a + b + c = 10$, substitute for c : $a + b + (b + 2) = 10$.

2. Write a in terms of b

This gives $a + 2b + 2 = 10$, so $a = 8 - 2b$.

3. List the possible b values

The integers a, b, c are all positive. Since $b > 0$ and $a = 8 - 2b > 0$, the only possible values are $b = 1, 2, 3$.

4. Exhaust the cases

If $b = 1$, then $a = 6$ and $c = 3$, so $abc = 18$. If $b = 2$, then $a = 4$ and $c = 4$, so $abc = 32$. If $b = 3$, then $a = 2$ and $c = 5$, so $abc = 30$.

5. Conclude the proof

The list contains every possible case, and in every case the product is even. Therefore abc is always even.

Final answer

The possible triples are $(6, 1, 3)$, $(4, 2, 4)$ and $(2, 3, 5)$, giving products 18, 32 and 30. Each product is even, so abc is always even.

Question 2

Binomial Expansion

2. A curve C has equation $y = f(x)$ where

$$f(x) = (2 - kx)^5$$

and k is a constant.

Given that when $f(x)$ is divided by $(4x - 5)$ the remainder is $\frac{243}{32}$

- (a) show that $k = \frac{2}{5}$

(2)

- (b) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{2}{5}x\right)^5$$

giving each term in simplest form.

(3)

Using the solution to part (b) and making your method clear,

- (c) find the gradient of C at the point where $x = 0$

(2)

Worked Solution - Question 2

1. Use the remainder theorem

Dividing by $4x - 5$ means the remainder is $f\left(\frac{5}{4}\right)$.

2. Substitute x equals five quarters

$$f\left(\frac{5}{4}\right) = \left(2 - \frac{5k}{4}\right)^5 = \frac{243}{32}.$$

3. Take the fifth root

Since $\frac{243}{32} = \left(\frac{3}{2}\right)^5$, we have $2 - \frac{5k}{4} = \frac{3}{2}$.

4. Find k

$$\frac{5k}{4} = \frac{1}{2}, \text{ so } k = \frac{2}{5}.$$

5. Expand the binomial

With $k = \frac{2}{5}$, $f(x) = \left(2 - \frac{2}{5}x\right)^5$.

6. Find the first three terms

$$2^5 + 5(2^4)\left(-\frac{2}{5}x\right) + 10(2^3)\left(-\frac{2}{5}x\right)^2 = 32 - 32x + \frac{64}{5}x^2.$$

7. Use the coefficient of x

For a polynomial, the gradient at $x = 0$ is the coefficient of x in the expansion.
Therefore $f'(0) = -32$.

Final answer

(a) $k = \frac{2}{5}$.

(b) $32 - 32x + \frac{64}{5}x^2$.

(c) -32 .

Question 3

Sequences & Series

3. A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_n = \cos^2\left(\frac{n\pi}{3}\right)$$

Find the exact values of

(a) (i) a_1

(ii) a_2

(iii) a_3

(3)

(b) Hence find the exact value of

$$\sum_{n=1}^{50} \left\{ n + \cos^2\left(\frac{n\pi}{3}\right) \right\}$$

You must make your method clear.

(4)

Worked Solution - Question 3

Topic group

1. Find the first term

$$a_1 = \cos^2\left(\frac{\pi}{3}\right) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}.$$

2. Find the second term

$$a_2 = \cos^2\left(\frac{2\pi}{3}\right) = \left(-\frac{1}{2}\right)^2 = \frac{1}{4}.$$

3. Find the third term

$$a_3 = \cos^2(\pi) = (-1)^2 = 1.$$

4. Recognise the repeating pattern

The values of $\cos^2\left(\frac{n\pi}{3}\right)$ repeat every 3 terms as $\frac{1}{4}, \frac{1}{4}, 1$.

5. Sum the n part

$$\sum_{n=1}^{50} n = \frac{50 \cdot 51}{2} = 1275.$$

6. Sum the cosine part

There are **16** complete groups of 3 terms in the first **48** terms, then the first 2 terms repeat. Hence the cosine-square sum is

$$16\left(\frac{1}{4} + \frac{1}{4} + 1\right) + \frac{1}{4} + \frac{1}{4} = 24 + \frac{1}{2} = \frac{49}{2}.$$

7. Add the two sums

$$\sum_{n=1}^{50} \left(n + \cos^2\left(\frac{n\pi}{3}\right)\right) = 1275 + \frac{49}{2} = \frac{2599}{2}.$$

Final answer

$$(a) \ a_1 = \frac{1}{4}, \ a_2 = \frac{1}{4}, \ a_3 = 1. \quad (b) \ \frac{2599}{2}.$$

Question 4

Laws of Logarithms

4. The weight of a baby mammal is monitored over a 16-month period.

The weight of the mammal, w kg, is given by

$$w = \log_a(t + 5) - \log_a 4 \qquad 2 \leq t \leq 18$$

where t is the age of the mammal in months and a is a constant.

Given that the weight of the mammal was 10 kg when $t = 3$

- (a) show that $a = 1.072$ correct to 3 decimal places. (3)

Using $a = 1.072$

- (b) find an equation for t in terms of w (3)

- (c) find the value of t when $w = 15$, giving your answer to 3 significant figures. (2)

Worked Solution - Question 4

1. Substitute the given information

When $t = 3$, $w = 10$. So $10 = \log_a(3 + 5) - \log_a 4 = \log_a 8 - \log_a 4$.

2. Combine the logarithms

$$\log_a 8 - \log_a 4 = \log_a \left(\frac{8}{4} \right) = \log_a 2, \text{ so } 10 = \log_a 2.$$

3. Find a

$10 = \log_a 2$ means $a^{10} = 2$, hence $a = 2^{1/10} = 1.071773 \dots = 1.072$ to 3 decimal places.

4. Start from the model

Using $a = 1.072$, $w = \log_{1.072}(t + 5) - \log_{1.072} 4$.

5. Combine the logs again

$$w = \log_{1.072} \left(\frac{t + 5}{4} \right).$$

6. Make t the subject

Changing from logarithmic form to exponential form gives $(1.072)^w = \frac{t + 5}{4}$, so $t = 4(1.072)^w - 5$.

7. Substitute w equals fifteen

When $w = 15$, $t = 4(1.072)^{15} - 5 = 6.3496 \dots$, so $t = 6.35$ months to 3 significant figures.

Final answer

(a) $a = 1.072$ to 3 d.p.

(b) $t = 4(1.072)^w - 5$.

(c) $t = 6.35$ months to 3 s.f.

Question 5

Trigonometric Equations

5.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$(3 \cos \theta - \tan \theta) \cos \theta = 2$$

can be written as

$$3 \sin^2 \theta + \sin \theta - 1 = 0 \quad (3)$$

(b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$$(3 \cos 2x - \tan 2x) \cos 2x = 2 \quad (5)$$

Worked Solution - Question 5

1. Rewrite the left side

$(3 \cos \theta - \tan \theta) \cos \theta = 3 \cos^2 \theta - \sin \theta$ because $\tan \theta \cos \theta = \sin \theta$.

2. Move everything to one side

The equation becomes $3 \cos^2 \theta - \sin \theta = 2$.

3. Use cos squared

Since $\cos^2 \theta = 1 - \sin^2 \theta$, $3(1 - \sin^2 \theta) - \sin \theta = 2$.

4. Show the required form

Expanding and rearranging gives $3 \sin^2 \theta + \sin \theta - 1 = 0$.

5. Apply the result to $2x$

For $(3 \cos 2x - \tan 2x) \cos 2x = 2$, take $\theta = 2x$. Then
 $3 \sin^2(2x) + \sin(2x) - 1 = 0$.

6. Solve the quadratic

Let $u = \sin(2x)$. Then $3u^2 + u - 1 = 0$, so $u = \frac{-1 \pm \sqrt{13}}{6}$.

7. Use the interval

Since $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$, we have $-\pi \leq 2x \leq \pi$. Solving $\sin(2x) = \frac{-1 + \sqrt{13}}{6}$
 and $\sin(2x) = \frac{-1 - \sqrt{13}}{6}$ in this interval gives
 $2x = -2.2665 \dots, -0.8751 \dots, 0.4492 \dots, 2.6924 \dots$

8. Divide by two

Therefore $x = -1.133 \dots, -0.4375 \dots, 0.2246 \dots, 1.346 \dots$, giving
 $x = -1.13, -0.438, 0.225, 1.35$ radians.

Final answer

(a) Proven.

(b) $x = -1.13, -0.438, 0.225, 1.35$ radians.

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7 marks

Question 6

Integration

6. The curve C_1 has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below, with the y values rounded to 4 decimal places where appropriate.

x	0	0.5	1	1.5	2
y	3	2.6833	2.4	2.1466	1.92

- (a) Use the trapezium rule with all the values of y in the table to find an approximation for

$$\int_0^2 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

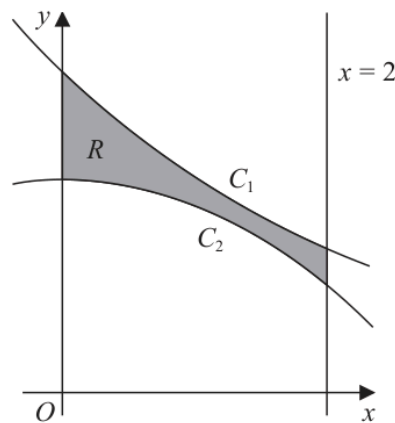


Figure 1

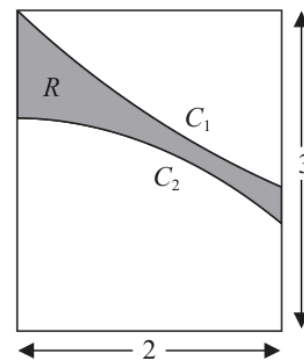


Figure 2

The region R , shown shaded in Figure 1, is bounded by

- the curve C_1
- the curve C_2 with equation $y = 2 - \frac{1}{4}x^2$
- the line with equation $x = 2$
- the y -axis

The region R forms part of the design for a logo shown in Figure 2.

The design consists of the shaded region R inside a rectangle of width 2 and height 3

Using calculus and the answer to part (a),

- (b) calculate an estimate for the percentage of the logo which is shaded.

(4)

Worked Solution - Question 6

1. Identify the strip width

The x values increase by 0.5 , so the trapezium-rule width is $h = 0.5$.

2. Apply the trapezium rule

$$\int_0^2 f(x) dx \approx \frac{0.5}{2} [3 + 1.92 + 2(2.6833 + 2.4 + 2.1466)].$$

3. Evaluate the estimate

This gives 4.84495 , so the approximation is 4.845 to 3 decimal places.

4. Find the area under C_2

$$\int_0^2 \left(2 - \frac{1}{4}x^2\right) dx = \left[2x - \frac{x^3}{12}\right]_0^2 = 4 - \frac{2}{3} = \frac{10}{3}.$$

5. Estimate the shaded area

The shaded region lies between C_1 and C_2 , so its estimated area is

$$4.845 - \frac{10}{3} = 1.5116\dots$$

6. Compare with the rectangle

The rectangle has area $2 \times 3 = 6$.

7. Convert to a percentage

$$\text{Percentage shaded} = \frac{1.5116\dots}{6} \times 100 = 25.19\dots\%, \text{ so the estimate is } 25.2\%$$

Final answer

(a) 4.845.

(b) 25.2%.

Question 7

Applications of Differentiation

7. The curve C has equation

$$y = \frac{12x^3(x - 7) + 14x(13x - 15)}{21\sqrt{x}} \quad x > 0$$

- (a) Write the equation of C in the form

$$y = ax^{\frac{7}{2}} + bx^{\frac{5}{2}} + cx^{\frac{3}{2}} + dx^{\frac{1}{2}}$$

where a , b , c and d are fully simplified constants.

(3)

The curve C has three turning points.

Using calculus,

- (b) show that the x coordinates of the three turning points satisfy the equation

$$2x^3 - 10x^2 + 13x - 5 = 0$$

(3)

Given that the x coordinate of one of the turning points is 1

- (c) find, using algebra, the exact x coordinates of the other two turning points.

(Solutions based entirely on calculator technology are not acceptable.)

(3)

Worked Solution - Question 7

Topic group

1. Expand the numerator

$$12x^3(x - 7) + 14x(13x - 15) = 12x^4 - 84x^3 + 182x^2 - 210x.$$

2. Divide by 21 root x

Since $\sqrt{x} = x^{1/2}$, divide each term by $21x^{1/2}$.

3. Write in the required form

$$y = \frac{12}{21}x^{7/2} - \frac{84}{21}x^{5/2} + \frac{182}{21}x^{3/2} - \frac{210}{21}x^{1/2} = \frac{4}{7}x^{7/2} - 4x^{5/2} + \frac{26}{3}x^{3/2} - 10x^{1/2}.$$

4. Differentiate

$$\frac{dy}{dx} = 2x^{5/2} - 10x^{3/2} + 13x^{1/2} - 5x^{-1/2}.$$

5. Factor the derivative

$$\frac{dy}{dx} = x^{-1/2}(2x^3 - 10x^2 + 13x - 5).$$

6. Use the turning point condition

At a turning point, $\frac{dy}{dx} = 0$. Since $x > 0$, $x^{-1/2} \neq 0$, so the x coordinates satisfy $2x^3 - 10x^2 + 13x - 5 = 0$.

7. Use the known root

One root is $x = 1$, so divide $2x^3 - 10x^2 + 13x - 5$ by $(x - 1)$.

8. Factor the cubic

$$2x^3 - 10x^2 + 13x - 5 = (x - 1)(2x^2 - 8x + 5).$$

9. Solve the quadratic

$$2x^2 - 8x + 5 = 0 \text{ gives } x = \frac{8 \pm \sqrt{64 - 40}}{4} = \frac{8 \pm 2\sqrt{6}}{4} = 2 \pm \frac{\sqrt{6}}{2}.$$

Final answer

$$(a) y = \frac{4}{7}x^{7/2} - 4x^{5/2} + \frac{26}{3}x^{3/2} - 10x^{1/2}.$$

(b) Proven.

$$(c) x = 2 - \frac{\sqrt{6}}{2} \text{ and } x = 2 + \frac{\sqrt{6}}{2}.$$

Question 8

Geometric Sequences

8. A geometric sequence has first term a and common ratio r

Given that $S_{\infty} = 3a$

- (a) show that $r = \frac{2}{3}$

(2)

Given also that

$$u_2 - u_4 = 16$$

where u_k is the k^{th} term of this sequence,

- (b) find the value of S_{10} giving your answer to one decimal place.

(5)

Worked Solution - Question 8

1. Use the sum to infinity formula

For a geometric sequence, $S_{\infty} = \frac{a}{1-r}$. The question gives $S_{\infty} = 3a$.

2. Find the common ratio

$\frac{a}{1-r} = 3a$. Since $u_2 - u_4 = 16$, $a \neq 0$, so $\frac{1}{1-r} = 3$ and $r = \frac{2}{3}$.

3. Use u_2 minus u_4

$$u_2 - u_4 = ar - ar^3 = ar(1 - r^2) = 16.$$

4. Substitute r

$$a \left(\frac{2}{3} \right) \left(1 - \frac{4}{9} \right) = 16.$$

5. Solve for a

$$a \cdot \frac{2}{3} \cdot \frac{5}{9} = 16, \text{ so } \frac{10a}{27} = 16 \text{ and } a = \frac{216}{5} = 43.2.$$

6. Find S_{10}

$$S_{10} = \frac{a(1-r^{10})}{1-r} = 3a \left(1 - \left(\frac{2}{3} \right)^{10} \right).$$

7. Evaluate

$$S_{10} = 3 \left(\frac{216}{5} \right) \left(1 - \left(\frac{2}{3} \right)^{10} \right) = \frac{92840}{729} = 127.352 \dots, \text{ so } S_{10} = 127 \text{ to 3 significant figures.}$$

Final answer

(a) $a = 43.2$.

(b) $S_{10} = 127.353\dots$, so **127** to 3 s.f.

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12 marks

Question 9

Circles

9.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

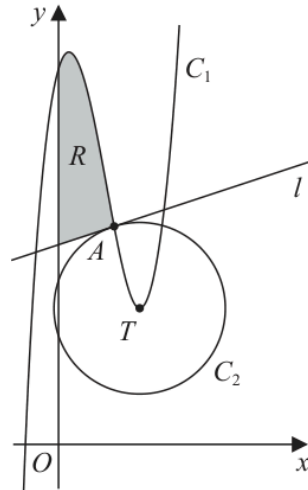


Figure 3

Figure 3 shows

- the curve C_1 with equation $y = x^3 - 5x^2 + 3x + 14$
- the circle C_2 with centre T

The point T is the minimum turning point of C_1

Using Figure 3 and calculus,

- (a) find the coordinates of T

(3)

The curve C_1 intersects the circle C_2 at the point A with x coordinate 2

- (b) Find an equation of the circle C_2

(3)

The line l shown in Figure 3, is the tangent to circle C_2 at A

- (c) Show that an equation of l is

$$y = \frac{1}{3}x + \frac{22}{3}$$

(3)

The region R , shown shaded in Figure 3, is bounded by C_1 , l and the y -axis.

- (d) Find the exact area of R .

(3)

Worked Solution - Question 9

Topic group

1. Differentiate the cubic

For $y = x^3 - 5x^2 + 3x + 14$, $\frac{dy}{dx} = 3x^2 - 10x + 3$.

2. Find stationary x values

$3x^2 - 10x + 3 = 0$ factors as $(3x - 1)(x - 3) = 0$, so $x = \frac{1}{3}$ or $x = 3$.

3. Identify the minimum

$\frac{d^2y}{dx^2} = 6x - 10$. At $x = 3$, this is positive, so the minimum turning point has $x = 3$.

4. Find T

Substitute $x = 3$: $y = 27 - 45 + 9 + 14 = 5$. Hence $T = (3, 5)$.

5. Find A

At $x = 2$ on C_1 , $y = 8 - 20 + 6 + 14 = 8$, so $A = (2, 8)$.

6. Find the circle radius

The centre of C_2 is $T = (3, 5)$ and it passes through $A = (2, 8)$. Thus $r^2 = (2 - 3)^2 + (8 - 5)^2 = 10$.

7. Write the circle equation

Therefore C_2 has equation $(x - 3)^2 + (y - 5)^2 = 10$.

8. Find the tangent gradient

The radius TA has gradient $\frac{8 - 5}{2 - 3} = -3$, so the tangent at A has gradient $\frac{1}{3}$.

9. Write the tangent equation

Using $A = (2, 8)$, $y - 8 = \frac{1}{3}(x - 2)$, so $y = \frac{1}{3}x + \frac{22}{3}$.

10. Set up the area

From $x = 0$ to $x = 2$, the cubic is above the line. The required area is

$$\int_0^2 \left[x^3 - 5x^2 + 3x + 14 - \left(\frac{1}{3}x + \frac{22}{3} \right) \right] dx.$$

11. Simplify the integrand

The integrand is $x^3 - 5x^2 + \frac{8}{3}x + \frac{20}{3}$.

12. Integrate

$$\left[\frac{x^4}{4} - \frac{5x^3}{3} + \frac{4x^2}{3} + \frac{20x}{3} \right]_0^2 = 4 - \frac{40}{3} + \frac{16}{3} + \frac{40}{3} = \frac{28}{3}.$$

Final answer

(a) $T = (3, 5)$.

(b) $(x - 3)^2 + (y - 5)^2 = 10$.

(c) $y = \frac{1}{3}x + \frac{22}{3}$.

(d) $\frac{28}{3}$.

Question 10

Laws of Logarithms

10. Given $a = \log_2 3$

(i) write, in simplest form, in terms of a ,

(a) $\log_2 9$

(b) $\log_2 \left(\frac{\sqrt{3}}{16} \right)$

(3)

(ii) Solve

$$3^x \times 2^{x+4} = 6$$

giving your answer, in simplest form, in terms of a .

(4)

Worked Solution - Question 10

1. Use the power rule for log 2 of 9

$$\log_2 9 = \log_2(3^2) = 2 \log_2 3 = 2a.$$

2. Split the second logarithm

$$\log_2 \left(\frac{\sqrt{3}}{16} \right) = \log_2(3^{1/2}) - \log_2(16).$$

3. Simplify in terms of a

$$\log_2(3^{1/2}) = \frac{1}{2} \log_2 3 = \frac{a}{2} \text{ and } \log_2(16) = 4, \text{ so the expression is } \frac{a}{2} - 4.$$

4. Take log base 2 of both sides

$$\text{For } 3^x \times 2^{x+4} = 6, \text{ take } \log_2 \text{ of both sides: } \log_2(3^x) + \log_2(2^{x+4}) = \log_2 6.$$

5. Rewrite each term

$$\log_2(3^x) = xa, \log_2(2^{x+4}) = x + 4, \text{ and } \log_2 6 = \log_2(2 \cdot 3) = 1 + a.$$

6. Solve for x

$$\text{So } xa + x + 4 = a + 1. \text{ Hence } x(a + 1) = a - 3, \text{ giving } x = \frac{a - 3}{a + 1}.$$

Final answer

$$(i)(a) \ 2a. \quad (i)(b) \ \frac{a}{2} - 4. \quad (ii) \ x = \frac{a - 3}{a + 1}.$$